

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250286227

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

SAKAMOTO; Shunya et al.

POWER STORAGE DEVICE

Abstract

A power storage device includes: an electrode including: a current collector having a surface; and a positive active material layer provided on the surface of the current collector; and a sealing member provided on the surface of the current collector, the sealing member surrounding the positive active material layer in a stacking direction. The current collector includes: a first metal foil including the surface; a second metal foil provided opposite to the positive active material layer with respect to first metal foil; and an adhesive layer provided between the first metal foil and the second metal foil, the adhesive layer bonding the first metal foil and the second metal foil with each other. The sealing member includes: a seal portion welded to the surface; and another seal portion welded to a side face of the current collector including a side face of the adhesive layer.

Inventors:	SAKAMOTO; Shunya (Kariya-shi, Aichi, JP), NAKAMURA; Tomohiro (Kariya-shi, Aichi, JP), HIROSE; Takayuki (Kariya-shi, Aichi, JP)
Applicant:	KABUSHIKI KAISHA TOYOTA JIDOSHOKKI (Kariya-shi, Aichi, JP)
Family ID:	88730178
Assignee:	KABUSHIKI KAISHA TOYOTA JIDOSHOKKI (Kariya-shi, Aichi, JP)
Appl. No.:	18/861417
Filed (or PCT Filed):	April 17, 2023
PCT No.:	PCT/JP2023/015333

Foreign Application Priority Data

JP	2022-078266	May. 11, 2022
----	-------------	---------------

Publication Classification

Int. Cl.: **H01M50/489** (20210101); **H01M50/46** (20210101); **H01M50/474** (20210101);
H01M50/477 (20210101); **H01M50/486** (20210101)

U.S. Cl.:

CPC **H01M50/489** (20210101); **H01M50/461** (20210101); **H01M50/474** (20210101);
H01M50/477 (20210101); **H01M50/486** (20210101);

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a power storage device.

BACKGROUND ART

[0002] Conventionally, a power storage device including a plurality of electrodes stacked in the stacking direction is known (see, for example, Patent Literature 1). Such an electrode as described above includes a positive current collector, a negative current collector, and an adhesive layer provided therebetween.

CITATION LIST

Patent Literature

[0003] Patent Literature 1: Japanese Unexamined Patent Publication No. 2005-317468

SUMMARY OF INVENTION

Technical Problem

[0004] Meanwhile, such a power storage device as described above further includes a sealing member for sealing the side faces of a plurality of electrodes, and the sealing member is welded to the surface of an electrode in some cases. In such a case, heat in the welding of the sealing member to the electrode is transferred to an adhesive layer, so that the adhesion strength by the adhesive layer is reduced. Accordingly, the electrode may be damaged.

[0005] An object of the present invention is to provide a power storage device that enables inhibition of damage to an electrode.

Solution to Problem

[0006] A power storage device according to the present disclosure includes: an electrode including: a current collector having a first main face; and an active material layer provided on the first main face; and a sealing member provided on a surface of the current collector, the sealing member surrounding the active material layer when viewed from a direction perpendicular to the first main face, in which the current collector includes: a first metal foil including the first main face; a second metal foil provided opposite to the active material layer with respect to the first metal foil; and an adhesive layer provided between the first metal foil and the second metal foil, the adhesive layer bonding the first metal foil and the second metal foil with each other, the sealing member includes: a first seal portion welded to the first main face; and a second seal portion welded to a side face of the current collector including a side face of the adhesive layer, and a heat-resistant temperature of the adhesive layer is higher than respective melting points of the first seal portion and the second seal portion.

[0007] In the power storage device, the current collector includes the adhesive layer that is provided between the first metal foil and the second metal foil and bonds the first metal foil and the second metal foil with each other. Further, the first seal portion is welded to the first main face of the first metal foil, and the second seal portion is welded to the side face of the current collector.

Here, the heat-resistant temperature of the adhesive layer of the current collector is higher than the respective melting points of the first seal portion and the second seal portion. Thus, even if heat in welding of the first seal portion and the second seal portion to the current collector is transferred to the adhesive layer, a decrease in adhesion strength by the adhesive layer is inhibited. As a result, damage to the current collector due to a decrease in adhesion strength by the adhesive layer is inhibited. Therefore, the power storage device enables inhibition of damage to the electrode.

[0008] The power storage device may further include a spacer, in which a plurality of the electrodes may be stacked in a stacking direction, the spacer may be provided between the plurality of the electrodes adjacent to each other in the stacking direction, the spacer may be welded to the sealing member, and the heat-resistant temperature of the adhesive layer may be higher than a melting point of the spacer. With this arrangement, even if heat in welding of the spacer to the sealing member is transferred to the adhesive layer, a decrease in adhesion strength by the adhesive layer is inhibited. Therefore, damage to the current collector due to a decrease in adhesion strength by the adhesive layer is inhibited, and as a result, damage to the electrodes is inhibited.

[0009] The current collector has a second main face opposite to the first main face, the sealing member may further include a third seal portion welded to the second main face, and the heat-resistant temperature of the adhesive layer may be higher than a melting point of the third seal portion. With this arrangement, even if heat in welding of the third seal portion to the current collector is transferred to the adhesive layer, a decrease in adhesion strength by the adhesive layer is inhibited. Therefore, damage to the current collector due to a decrease in adhesion strength by the adhesive layer is inhibited, and as a result, damage to the electrodes is inhibited.

[0010] The adhesive layer may contain a thermosetting resin, and the sealing member may contain a thermoplastic resin. Accordingly, use of the thermosetting resin as a component of the adhesive layer results in an increase of the heat-resistant temperature of the adhesive layer, and use of the thermoplastic resin as a component of the sealing member results in an increase of sealing performance by the sealing member.

[0011] The current collector may further include a carbon coat layer provided at least either between the first metal foil and the adhesive layer or between the second metal foil and the adhesive layer. With this arrangement, the adhesion strength between the first metal foil and the adhesive layer and the adhesion strength between the second metal foil and the adhesive layer can be reliably secured.

Advantageous Effects of Invention

[0012] According to the present disclosure, provided can be a power storage device that enables inhibition of damage to an electrode.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0013] FIG. 1 is a schematic sectional view of a power storage device according to an embodiment.

[0014] FIG. 2 is a partially enlarged view of FIG. 1.

[0015] FIG. 3 is a view of steps of a method for manufacturing the power storage device illustrated in FIG. 1.

[0016] FIG. 4 is a view of steps of the method for manufacturing the power storage device illustrated in FIG. 1.

[0017] FIG. 5 is view of steps of the method for manufacturing the power storage device illustrated in FIG. 1.

[0018] FIG. 6 is a view of steps of the method for manufacturing the power storage device illustrated in FIG. 1.

[0019] FIG. 7 is a partially enlarged view of a power storage device according to a modification.

DESCRIPTION OF EMBODIMENTS

[0020] Hereinafter, embodiments of the present disclosure will be described in detail with reference to the drawings. Note that in the drawings, the same or corresponding parts are denoted by the same reference signs, and the redundant description will be omitted.

[0021] A power storage device **1** illustrated in FIG. **1** is, for example, a power storage module to be used for a battery of a forklift, a hybrid vehicle, or an electric vehicle. The power storage device **1** is, for example, a secondary battery such as a nickel-metal hydride secondary battery or a lithium ion secondary battery. In the present embodiment, the power storage device **1** is a lithium ion secondary battery.

[0022] As illustrated in FIG. **1**, the power storage device **1** includes a stack **2** and a seal **3**. The stack **2** includes a plurality of bipolar electrodes **21** stacked in the stacking direction **D**, a positive terminal electrode **22**, and a negative terminal electrode **23**. The stack **2** is formed by stacking the plurality of bipolar electrodes **21** between the positive terminal electrode **22** and the negative terminal electrode **23**. Such a bipolar electrode **21** as described above includes a current collector **24**, a positive active material layer **25**, and a negative active material layer **26**.

[0023] The current collector **24** is, for example, rectangular when viewed from the stacking direction **D**. The current collector **24** has a first main face **24a** as one surface and a second main face **24b** as the other surface opposite to the first main face **24a**. That is, the current collector **24** has the first main face **24a** and the second main face **24b** opposite to each other in the stacking direction **D**. The positive active material layer **25** is provided on the first main face **24a** of the current collector **24**. The positive active material layer **25** is, for example, rectangular when viewed from the stacking direction **D**. The first main face **24a** of the current collector **24** includes a non-coated region where the positive active material layer **25** is not provided. The non-coated region surrounds the positive active material layer **25** when viewed from the stacking direction **D**.

[0024] The negative active material layer **26** is provided on the second main face **24b** of the current collector **24**. The negative active material layer **26** is, for example, rectangular when viewed from the stacking direction **D**. The second main face **24b** of the current collector **24** includes a non-coated region where the negative active material layer **26** is not provided. The non-coated region surrounds the negative active material layer **26** when viewed from the stacking direction **D**. The plurality of bipolar electrodes **21** is stacked such that the positive active material layer **25** of one bipolar electrode **21** and the negative active material layer **26** of the other bipolar electrode **21** adjacent to each other in the stacking direction **D** are opposite to each other. That is, the plurality of bipolar electrodes **21** is stacked such that, of the bipolar electrodes **21** adjacent to each other in the stacking direction **D**, the first main face **24a** of the current collector **24** of one bipolar electrode **21** and the second main face **24b** of the current collector **24** of the other bipolar electrode **21** are opposite to each other.

[0025] The positive terminal electrode **22** is stacked and disposed at one end of the plurality of bipolar electrodes **21** in the stacking direction **D**. The positive terminal electrode **22** includes a current collector **24** and a positive active material layer **25** provided on a first main face **24a** that the current collector **24** has. The positive terminal electrode **22** is different from such a bipolar electrode **21** as described above in that a negative active material layer **26** is not provided on a second main face **24b** that the current collector **24** has. Note that a positive active material layer **25** or a negative active material layer **26** is not provided on the second main face **24b** of the current collector **24** of the positive terminal electrode **22**. The second main face **24b** of the current collector **24** of the positive terminal electrode **22** serves as a positive terminal face that the power storage device **1** has. Other configurations of the positive terminal electrode **22** may be the same as those of the bipolar electrode **21**. The positive terminal electrode **22** is disposed such that the positive active material layer **25** of the positive terminal electrode **22** is opposite to the negative active material layer **26** of the bipolar electrode **21**. That is, the positive terminal electrode **22** is stacked such that the first main face **24a** of the current collector **24** of the positive terminal electrode **22** and the

second main face **24b** of the current collector **24** of the bipolar electrode **21** adjacent to the positive terminal electrode **22** in the stacking direction D are opposite to each other.

[0026] The negative terminal electrode **23** is disposed opposite to the positive terminal electrode **22** with respect to the plurality of bipolar electrodes **21**. That is, the positive terminal electrode **22** is stacked and disposed at the one end of the plurality of bipolar electrodes **21** in the stacking direction D, and the negative terminal electrode **23** is stacked and disposed at the other end of the plurality of bipolar electrodes **21** in the stacking direction D. The negative terminal electrode **23** includes a current collector **24** and a negative active material layer **26** provided on a second main face **24b** that the current collector **24** has. The negative terminal electrode **23** is different from such a bipolar electrode **21** as described above in that a positive active material layer **25** is not provided on a first main face **24a** that the current collector **24** has. Note that a positive active material layer **25** or a negative active material layer **26** is not provided on the first main face **24a** of the current collector **24** of the negative terminal electrode **23**. The first main face **24a** of the current collector **24** of the negative terminal electrode **23** serves as a negative terminal face that the power storage device **1** has. Other configurations of the negative terminal electrode **23** may be the same as those of the bipolar electrode **21**. The negative terminal electrode **23** is disposed such that the negative active material layer **26** of the negative terminal electrode **23** is opposite to the positive active material layer **25** of the bipolar electrode **21**. That is, the negative terminal electrode **23** is stacked such that the second main face **24b** of the current collector **24** of the negative terminal electrode **23** and the first main face **24a** of the current collector **24** of the bipolar electrode **21** adjacent to the negative terminal electrode **23** in the stacking direction D are opposite to each other. Each internal space S containing an electrolytic solution is formed between the bipolar electrodes **21** adjacent to each other, between the bipolar electrode **21** and the positive terminal electrode **22** adjacent to each other, and between the bipolar electrode **21** and the negative terminal electrode **23** adjacent to each other in the stacking direction D.

[0027] The stack **2** includes a plurality of separators **27**. Each of the separators **27** is disposed between the bipolar electrodes **21** adjacent to each other, between the bipolar electrode **21** and the positive terminal electrode **22** adjacent to each other, and between the bipolar electrode **21** and the negative terminal electrode **23** adjacent to in the stacking direction D. Such a separator **27** as described above is located between the positive active material layer **25** and the negative active material layer **26** opposite to each other. The separator **27** is, for example, sheet-like. The separator **27** is, for example, rectangular when viewed from the stacking direction D. When viewed from the stacking direction D, the separator **27** has an outer edge located outside an outer edge that the positive active material layer **25** has and an outer edge that the negative active material layer **26** has. The separator **27** has a peripheral edge located between a sealing member **4** and a spacer **5** described later. The separator **27** is a member that allows a charge carrier such as a lithium ion to pass therethrough. Each of such separators **27** as described above isolates the bipolar electrodes **21** adjacent to each other, the bipolar electrode **21** and the positive terminal electrode **22** adjacent to each other, and the bipolar electrode **21** and the negative terminal electrode **23** adjacent each other in the stacking direction D. This arrangement prevents an electrical short circuit due to contact between the bipolar electrodes **21** adjacent to each other, between the bipolar electrode **21** and the positive terminal electrode **22** adjacent each other, and between the bipolar electrode **21** and the negative terminal electrode **23** adjacent each other in the stacking direction D.

[0028] The current collector **24** is a chemically inert electric conductor for continuously flowing a current to the positive active material layer **25** and the negative active material layer **26** during discharge or charge of the power storage device **1**. The current collector **24** may be, for example, plate-like, foil-like, sheet-like, film-like, or mesh-like, for example. When the current collector **24** is foil-like, the thickness of the current collector **24** may be within a range of 1 μm or more and 100 μm or less. A coated layer may be formed on the surface of the current collector **24** by a known method such as plating or spray coating. Note that for example, among such current collectors **24**

as described above of the bipolar electrodes **21**, the positive terminal electrode **22**, and the negative terminal electrode **23**, some of the current collectors **24** may have a thickness of 100 μm or more. In this case, the structural stability of the stack **2** is enhanced.

[0029] The positive active material layer **25** contains a positive active material capable of occluding and releasing a charge carrier such as a lithium ion. Examples of the positive active material include a composite oxide, lithium metal, and sulfur. The composition of the composite oxide includes, for example, at least one of iron, manganese, titanium, nickel, cobalt, and aluminum, and lithium. Examples of the composite oxide include olivine type lithium iron phosphate (LiFePO_4), LiCoO_2 , and LiNiMnCoO_2 .

[0030] The negative active material layer **26** contains a negative active material capable of occluding and releasing a charge carrier such as a lithium ion. Examples of the negative active material include carbon such as graphite, artificial graphite, highly oriented graphite, mesocarbon microbeads, hard carbon, and soft carbon; metal compounds; elements capable of being alloyed with lithium or compounds thereof; and boron-added carbon. Examples of the elements capable of being alloyed with lithium include silicon (silicon) and tin.

[0031] Each of the positive active material layer **25** and the negative active material layer **26** may contain a binder and a conductive auxiliary in addition to such an active material as described above. The binder plays a role of connecting the active material or the conductive auxiliary to each other and maintaining the conductive network in the electrode. Examples of the binder include fluorine-containing resins such as polyvinylidene difluoride, polytetrafluoroethylene, and fluoroelastomer; thermoplastic resins such as polypropylene and polyethylene; imide resins such as polyimide and polyamideimide; alkoxysilyl group-containing resins; acrylic resins such as polyacrylic acid and polymethacrylic acid; styrene-butadiene rubber; carboxymethyl cellulose; alginates such as sodium alginate and ammonium alginate; water-soluble cellulose ester crosslinked products; and starch-acrylic acid graft polymers. These binders can be used alone or in combination. The conductive auxiliary is, for example, a conductive material such as acetylene black, carbon black, or graphite, and can enhance electrical conductivity. As a viscosity modifying solvent, N-methyl-2-pyrrolidone is used, for example.

[0032] For formation of the positive active material layer **25** on the first main face **24a** and formation of the negative active material layer **26** on the second main face **24b**, for example, used is a known method such as roll coating, die coating, dip coating, doctor blading, spray coating, or curtain coating. Specifically, an active material, a solvent, and if necessary, a binder and a conductive auxiliary are mixed to produce a slurry-like active material layer forming composition, the active material layer forming composition is applied to the first main face **24a** or the second main face **24b**, and then the applied active material layer forming composition is dried to obtain the positive active material layer **25** or the negative active material layer **26**. The solvent is, for example, N-methyl-2-pyrrolidone, methanol, methyl isobutyl ketone, or water. In order to increase the electrode density, the dried positive active material layer **25** or the dried negative active material layer **26** may be further compressed.

[0033] The separator **27** is, for example, a porous sheet or a nonwoven fabric containing a polymer that absorbs and holds an electrolyte. The material of the separator **27** is, for example, polypropylene, polyethylene, polyolefin, or polyester. The separator **27** may have a single layer structure or a multilayer structure. When the separator **27** has a multilayer structure, the separator **27** may include, for example, a base layer and a pair of adhesive layers, and may be adhesively fixed to the positive active material layer **25** and the negative active material layer **26** by the pair of adhesive layers. The separator **27** may include a ceramic layer serving as a heat resistant layer. The separator **27** may be reinforced with a vinylidene difluoride resin compound.

[0034] Examples of the electrolyte with which the separator **27** is impregnated include a liquid electrolyte (electrolytic solution) containing a nonaqueous solvent and an electrolyte salt dissolved in the nonaqueous solvent, or a polymer gel electrolyte containing an electrolyte held in a polymer

matrix. In a case where the separator **13** is impregnated with an electrolyte, a known lithium salt such as LiClO_4 , LiAsF_6 , LiPF_6 , LiBF_4 , LiCF_3SO_3 , $\text{LiN}(\text{FSO}_2)_2$, or $\text{LiN}(\text{CF}_3\text{SO}_2)_2$ can be used as the electrolyte salt. As the nonaqueous solvent, a known solvent such as cyclic carbonates, cyclic esters, chain carbonates, chain esters, or ethers can be used. Note that two or more of these known solvent materials may be used in combination. In the present embodiment, the electrolyte with which the separator **27** is impregnated is an electrolytic solution.

[0035] The seal **3** is a member that seals the internal spaces **S**. The seal **3** is provided on a side face of the stack **2**. The seal **3** seals the side face of the stack **2**. The seal **3** has, for example, a rectangular cylindrical shape. The seal **3** is electrically insulative. The seal **3** includes a plurality of the sealing members **4**, a plurality of the spacers **5**, and a welded portion **6**.

[0036] Each sealing member **4** is provided on the surface of the corresponding current collector **24** and surrounds the corresponding positive active material layer **25** and the corresponding negative active material layer **26** when viewed from the stacking direction **D** (direction perpendicular to the first main face **24a**). Each spacer **5** is provided between the bipolar electrodes **21** adjacent to each other, between the bipolar electrode **21** and the positive terminal electrode **22** adjacent to each other, and between the bipolar electrode **21** and the negative terminal electrode **23** adjacent each other in the stacking direction **D**. The spacer **5** has, for example, a rectangular frame shape when viewed from the stacking direction **D**. The welded portion **6** has, for example, a rectangular cylindrical shape. The welded portion **6** reaches both ends of the stack **2** in the stacking direction **D**. The welded portion **6** is integrated by welding the respective outer edges of the sealing members **4** and the spacers **5**.

[0037] Hereinafter, a description will be given focusing on a single electrode (e.g., a single bipolar electrode **21**) with reference to FIG. **2**. As illustrated in FIG. **2**, a current collector **24** includes a first metal foil **241**, a second metal foil **242**, and an adhesive layer **243**. The first metal foil **241** has one surface as a first main face **24a**. That is, a positive active material layer **25** is provided on the one surface of the first metal foil **241** as the first main face **24a** of the current collector **24**. The first metal foil **241** is, for example, rectangular when viewed from the stacking direction **D**. The first metal foil **241** is, for example, an aluminum foil. The first metal foil **241** may be, for example, a nickel foil, a titanium foil, or a stainless steel foil. The first metal foil **241** may be an alloy foil or a clad foil of two or more metals among the metals described above.

[0038] The second metal foil **242** is provided opposite to the positive active material layer **25** with respect to first metal foil **241**. The second metal foil **242** has one surface as a second main face **24b**. That is, a negative active material layer **26** is provided on the one surface of the second metal foil **242** as the second main face **24b** of the current collector **24**. The second metal foil **242** is, for example, rectangular when viewed from the stacking direction **D**. When viewed from stacking direction **D**, a side face (outer edge) **242c** that the second metal foil **242** has is substantially aligned with a side face (outer edge) **241c** that the first metal foil **241** has. In the present embodiment, the second metal foil **242** is, for example, a copper foil. The second metal foil **242** may be, for example, a nickel foil, a titanium foil, or a stainless steel foil. The second metal foil **242** may be an alloy foil or a clad foil of two or more metals among the metals described above.

[0039] The adhesive layer **243** is provided between the first metal foil **241** and the second metal foil **242**. The adhesive layer **243** is, for example, rectangular when viewed from the stacking direction **D**. When viewed from the stacking direction **D**, a side face (outer edge) **243c** that the adhesive layer **243** has is substantially aligned with the side face **241c** of the first metal foil **241** and the side face **242c** of the second metal foil **242**. The side face **241c**, the side face **242c**, and the side face **243c** are constituents of a side face (outer edge) **24c** that the current collector **24** has.

[0040] The adhesive layer **243** bonds the first metal foil **241** and the second metal foil **242** with each other. The adhesive layer **243** is joined to the first metal foil **241** and the second metal foil **242**. The adhesive layer **243** is conductive. The adhesive layer **243** contains, for example, a

conductive resin material. Examples of the conductive resin material include a conductive polymer material or a resin resulting from adding of a conductive filler to a non-conductive polymer material.

[0041] In the present embodiment, the adhesive layer **243** contains an adhesive component and a conductive component. The adhesive component functions as an adhesive for bonding the first metal foil **241** and the second metal foil **242** with each other. The adhesive component is, for example, a thermosetting resin. The adhesive component contains, for example, a resin and a curing agent mixed with the resin. The resin of the adhesive component is, for example, an olefin-based resin. Specifically, the resin of the adhesive component is, for example, polypropylene or polyethylene. The curing agent of the adhesive component is, for example, an isocyanate-based curing agent or an epoxy-based curing agent. In order to join two metal foils by an adhesive layer containing an olefin-based resin as an adhesive component, an epoxy-based curing agent that generates less gas due to a curing reaction of the adhesive component can be suitably used. The conductive component electrically connects the first metal foil **241** and the second metal foil **242**. The conductive component is, for example, conductive particles dispersed in the adhesive component. The conductive particles are, for example, graphite particles or metal particles. As the conductive particles, for example, spherical particles each having a metal film formed on the surface of a core particle such as a resin or ceramic can also be used.

[0042] The sealing member **4** includes a first seal portion **41**, a third seal portion **42**, and a second seal portion **43**. The first seal portion **41** is provided on the first main face **24a**. The first seal portion **41** has, for example, a rectangular frame shape when viewed from the stacking direction D. The first seal portion **41** surrounds the positive active material layer **25** and the negative active material layer **26** when viewed from the stacking direction D. The first seal portion **41** is provided in a non-coated region on the first main face **24a** of the current collector **24**. The first seal portion **41** has an outer edge **41c** substantially aligned with the side face **24c** of the current collector **24** when viewed from the stacking direction D. The first seal portion **41** has an inner edge **41d** separated from the positive active material layer **25**. When viewed from the stacking direction D, the first main face **24a** of the current collector **24** is exposed to the internal space S between the positive active material layer **25** and the inner edge **41d** of the first seal portion **41**. The first seal portion **41** is welded to the first main face **24a**.

[0043] The third seal portion **42** is provided on the second main face **24b**. The third seal portion **42** has, for example, a rectangular frame shape when viewed from the stacking direction D. The third seal portion **42** surrounds the positive active material layer **25** and the negative active material layer **26** when viewed from the stacking direction D. The third seal portion **42** is provided in a non-coated region on the second main face **24b** of the current collector **24**. The third seal portion **42** has an outer edge **42c** substantially aligned with the side face **24c** of the current collector **24** when viewed from the stacking direction D. The third seal portion **42** has an inner edge **42d** separated from the negative active material layer **26**. When viewed from the stacking direction D, the second main face **24b** of the current collector **24** is exposed to the internal space S between the negative active material layer **26** and the third seal portion **42**. The inner edge **42d** of the third seal portion **42** is substantially aligned with the inner edge **41d** of the first seal portion **41** when viewed from the stacking direction D. The third seal portion **42** is substantially the same in thickness as the first seal portion **41**. The third seal portion **42** is welded to the second main face **24b**.

[0044] The second seal portion **43** is provided on the side face **24c** of the current collector **24**. The second seal portion **43** is located outside the side face **24c** of the current collector **24**, the outer edge **41c** of the first seal portion **41**, and the outer edge **42c** of the third seal portion **42**. The second seal portion **43** is connected to the first seal portion **41** and the third seal portion **42**. The second seal portion **43** covers the side face **24c** of the current collector **24**. The second seal portion **43** covers the side face **241c** of the first metal foil **241**, the side face **242c** of the second metal foil **242**, and the side face **243c** of the adhesive layer **243**.

[0045] The second seal portion **43** may be welded to the side face **24c** of the current collector **24**. For example, the second seal portion **43** may be welded to the side face **241c** of the first metal foil **241**, the side face **242c** of the second metal foil **242**, and the side face **243c** of the adhesive layer **243**. The second seal portion **43** seals the side face **24c** of the current collector **24**. Each of the first seal portion **41**, the third seal portion **42**, and the second seal portion **43** is a partial region of a sealing member **4** integrally formed of the same material.

[0046] A spacer **5** is provided between such sealing members **4** as described above adjacent to each other in the stacking direction **D**. The spacer **5** has, for example, a rectangular frame shape. An inner edge **5d** that the rectangular frame-shaped spacer **5** has may be farther from the positive active material layer **25** or the negative active material layer **26** than each of the inner edge **41d** of the rectangular frame-shaped first seal portion **41** and the inner edge **42d** of the rectangular frame-shaped third seal portion **42** is when viewed from the stacking direction **D**. In the present embodiment, when viewed from the stacking direction **D**, the inner edge **41d** of the first seal portion **41** is located between the positive active material layer **25** and the inner edge **5d** of the spacer **5**, and the inner edge **42d** of the third seal portion **42** is located between the negative active material layer **26** and the inner edge **5d** of the spacer **5**. The spacer **5** is different in member from such a sealing member **4** as described above. Note that the inner edge **5d** of the spacer **5** may be closer to the positive active material layer **25** or the negative active material layer **26** than each of the inner edge **41d** of the first seal portion **41** and the inner edge **42d** of the third seal portion **42** is when viewed from the stacking direction **D**.

[0047] The spacer **5** is sandwiched between a pair of such sealing members **4** as described above adjacent to each other in the stacking direction **D**. The spacer **5** is in contact with the sealing members **4**. The spacer **5** forms an internal space **S** for containing an electrolytic solution, with each other with a pair of such bipolar electrodes **21** as described above adjacent to each other in the stacking direction **D** and the pair of sealing members **4** adjacent to each other in the stacking direction **D**. Note that a filling port for filling the electrolytic solution into the internal space **S** is provided to the spacer **5**.

[0048] The thickness of the spacer **5** in the stacking direction **D** may be larger than the thickness of each of the first seal portion **41** and the third seal portion **42**. The thickness of the spacer **5** is, for example, twice or more the thickness of the first seal portion **41** or the thickness of the third seal portion **42**. That is, the thickness of the spacer **5** may be smaller than the sum of the thickness of the first metal foil **241**, the thickness of the second metal foil **242**, the thickness of the adhesive layer **243**, the thickness of the first seal portion **41**, and the thickness of the third seal portion **42**.

[0049] The spacer **5** is welded to such a sealing member **4** as described above on the outer side (welded portion **6**) of the side face **24c** of the current collector **24**. Specifically, the respective outer edges of such sealing members **4** as described above and such spacers **5** as described above are melted and then resolidified to form the welded portion **6**. That is, the outer edges of the sealing members **4** and the spacer **5** are mutually welded to form the welded portion **6**. That is, in the welded portion **6**, a region having a predetermined width from the outer edge of each spacer **5** is welded to a region having a predetermined width from the outer edge of the corresponding sealing member **4**. In the welded portion **6**, the spacer **5** and the sealing member **4** are compatible with each other.

[0050] The outer edge of the sealing member **4** is a portion of the sealing member **4** that is located opposite to the current collector **24** and is closer to the second seal portion **43** than the current collector **24** is. The outer edge of the sealing member **4** is a portion to be welded between the sealing member **4** and the spacer **5**. The outer edge of the spacer **5** is a portion of the spacer **5** that overlaps the outer edge of the sealing member **4** when viewed from the stacking direction **D**. The outer edge of the spacer **5** is a portion to be welded between the sealing member **4** and the spacer **5**. In the present embodiment, the welded portion **6** does not reach the side face **24c** of the current collector **24**. The welded portion **6** may reach the side face **24c** of the current collector **24**. Note

that a filling port in communication with the filling port of the spacer 5 is provided to the welded portion 6.

[0051] The sealing member 4 and the spacer 5 each contain a thermoplastic resin. The material of each of the sealing member 4 and the spacer 5 is, for example, acid-modified polyethylene (acid-modified PE), acid-modified polypropylene (acid-modified PP), polyethylene, or polypropylene. The sealing member 4 and the spacer 5 are each electrolyte resistant. The materials of the sealing member 4 and the spacer 5 may be the same or different from each other. In the present embodiment, the material of the sealing member 4 is, for example, acid-modified polyethylene or acid-modified polypropylene. In the present embodiment, the material of the spacer 5 is, for example, polyethylene or polypropylene.

[0052] Acid-modified polyethylene and acid-modified polypropylene are easily bonded to metal as compared with non-acid-modified polyethylene and non-acid-modified polypropylene. That is, acid-modified polyethylene and acid-modified polypropylene are resin materials higher in adhesiveness to metal than non-acid-modified polyethylene and non-acid-modified polypropylene. The material of each of the first metal foil 241 and the second metal foil 242 of the current collector 24 is metal. Thus, the first seal portion 41, the third seal portion 42, and the second seal portion 43 each contain acid-modified polyethylene or acid-modified polypropylene, resulting in improvement of the joining strength of each of the first seal portion 41, the third seal portion 42, and the second seal portion 43 to the first metal foil 241 or the second metal foil 242.

[0053] The heat-resistant temperature of the adhesive layer 243 (hereinafter, may be simply referred to as “heat-resistant temperature”) is higher than the respective melting points of the first seal portion 41, the third seal portion 42, the second seal portion 43, and the spacer 5. Specifically, the heat-resistant temperature of the adhesive component of the adhesive layer 243 is higher than the respective melting points of the first seal portion 41, the third seal portion 42, the second seal portion 43, and the spacer 5. The melting points of the first seal portion 41, the third seal portion 42, the second seal portion 43, and the spacer 5 are each lower than 150 degrees, for example. The heat-resistant temperature of the adhesive layer 243 is, for example, 150 degrees or higher.

[0054] The heat-resistant temperature of the adhesive layer 243 is a limit temperature at which a favorable adhesion state between the first metal foil 241 and the second metal foil 242 is maintained. Specifically, the heat-resistant temperature of the adhesive layer 243 is a limit temperature at which the adhesion strength between the first metal foil 241 and the second metal foil 242 by the adhesive layer 243 is a predetermined threshold or larger. The predetermined threshold may be, for example, 1% of the adhesion strength of the adhesive layer 243 at room temperature. In typical, as the temperature of the adhesive layer 243 is increased by heating, the adhesion strength of the adhesive layer 243 tends to be lower than that before heating. As the temperature of the adhesive layer 243 becomes higher than the heat-resistant temperature, the adhesion strength of the adhesive layer 243 decreases to a value lower than the predetermined threshold. When the adhesion strength of the adhesive layer 243 is smaller than the predetermined threshold, a favorable adhesion state between the first metal foil 241 and the second metal foil 242 is not maintained, so that the current collector 24 may be damaged. In the present embodiment, the heat-resistant temperature of the adhesive layer 243 is higher than the melting points of the first seal portion 41, the third seal portion 42, the second seal portion 43, and the spacer 5. Thus, in welding of the first seal portion 41, the third seal portion 42, the second seal portion 43, or the spacer 5, even if the temperature of the adhesive layer 243 reaches the melting point of the first seal portion 41, the melting point of the third seal portion 42, the melting point of the second seal portion 43, or the melting point of the spacer 5, the current collector 24 is hardly damaged. Examples of the damage of the current collector 24 include peeling of the first metal foil 241 or the second metal foil 242, a change in the relative position between the first metal foil 241 and the second metal foil 242 (positional displacement between the first metal foil 241 and the second metal foil 242), and deformation (e.g., wrinkle) of the first metal foil 241 or the second metal foil

242.

[0055] In the present embodiment, the adhesion strength of the adhesive layer **243** heated to the heat-resistant temperature (hereinafter, referred to as “high-temperature strength”) is 1% or larger of the adhesion strength of the adhesive layer **243** at room temperature (hereinafter, referred to as “room-temperature strength”). When the adhesion strength of the heated adhesive layer **243** is 1% or larger of the room-temperature strength, a favorable adhesion state between the first metal foil **241** and the second metal foil **242** by the adhesive layer **243** is maintained. That is, when the temperature of the adhesive layer **243** is equal to or lower than the heat-resistant temperature, the adhesion strength of the adhesive layer **243** is 1% or larger of the room-temperature strength, and thus the favorable adhesion state between the first metal foil **241** and the second metal foil **242** is maintained. In the present embodiment, the heat-resistant temperature of the adhesive layer **243** is higher than the melting points of the first seal portion **41**, the third seal portion **42**, the second seal portion **43**, and the spacer **5**. Thus, in welding of the first seal portion **41**, the third seal portion **42**, the second seal portion **43**, or the spacer **5**, even if the temperature of the adhesive layer **243** reaches the melting point of the first seal portion **41**, the melting point of the third seal portion **42**, the melting point of the second seal portion **43**, or the melting point of the spacer **5**, the adhesion strength of the adhesive layer **243** is 1% or larger of the room-temperature strength. That is, even if the temperature of the adhesive layer **243** reaches the melting point of the first seal portion **41**, the melting point of the third seal portion **42**, the melting point of the second seal portion **43**, or the melting point of the spacer **5**, a favorable adhesion state between the first metal foil **241** and the second metal foil **242** by the adhesive layer **243** is maintained, so that the current collector **24** is hardly damaged.

[0056] The room-temperature strength is measured, for example, as follows. That is, first, a current collector **24** (a current collector **24** provided with none of a positive active material layer **25**, a negative active material layer **26**, and a sealing member **4**) is prepared. Next, at room temperature, for example, by a 180-degree peeling strength test (in accordance with JIS Z 0237:2009 as an example), the peeling strength when a first metal foil **241** is peeled off from a second metal foil **242** is measured as a room-temperature strength. The high-temperature strength is measured, for example, as follows. That is, first, similarly to the measurement of the room-temperature strength, a current collector **24** is prepared. Next, the current collector **24** is heated to the heat-resistant temperature of an adhesive layer **243**. Next, with the temperature of the current collector **24** heated to the heat-resistant temperature of the adhesive layer **243** maintained, the peeling strength when a first metal foil **241** is peeled off from a second metal foil **242** is measured as a high-temperature strength, similarly to the measurement of the room-temperature strength.

[0057] Next, a method for manufacturing a power storage device **1** will be described. First, as illustrated in FIG. **3**, a bipolar electrode **21**, a first sealing member **71**, and a second sealing member **72** are prepared. Next, the first sealing member **71** is disposed on a first main face **24a** that a current collector **24** has, and the second sealing member **72** is disposed on a second main face **24b** that the current collector **24** has. The first sealing member **71** and the second sealing member **72** are each disposed so as to protrude from a side face **24c** that the current collector **24** has. That is, when viewed from the stacking direction **D**, the first sealing member **71** disposed on the first main face **24a** of the current collector **24** and the second sealing member **72** disposed on the second main face **24b** of the current collector **24** each have a portion overlapping the current collector **24** and a portion not overlapping the current collector **24**. When each of the first sealing member **71** and the second sealing member **72** is disposed so as to extend along the corresponding side of the current collector **24**, the first sealing member **71** and the second sealing member **72** may be temporarily secured to the current collector **24** by spot welding using ultrasonic waves or heat, for example.

[0058] Next, as illustrated in FIG. **4**, the first sealing member **71** and the second sealing member **72** are each heated by a pair of heaters **73**. Further, the first sealing member **71** and the second sealing member **72** are each pressurized toward the current collector **24** by the pair of heaters **73**. The pair

of heaters 73 is, for example, an impulse sealer. Heating of the first sealing member 71 and the second sealing member 72 is performed with the current collector 24 sandwiched between the pair of heaters 73 through the first sealing member 71 and the second sealing member 72. The first sealing member 71 is heated until being melted. Similarly, the second sealing member 72 is heated until being melted. The first sealing member 71 and the second sealing member 72 heated until being melted are pressurized toward the current collector 24 by the pair of heaters 73. Thereafter, in response to stop of the heating by the pair of heaters 73, the first sealing member 71 and the second sealing member 72 are cooled, and accordingly, the first sealing member 71 and the second sealing member 72 are welded to the current collector 24. As a result, a portion of the first sealing member 71 overlapping the first main face 24a is welded to the first main face 24a to form a first seal portion 41, and a portion of the second sealing member 72 overlapping the second main face 24b is welded to the second main face 24b to form a third seal portion 42. Further, a portion of the first sealing member 71 located outside the side face 24c of the current collector 24 and a portion of the second sealing member 72 located outside the side face 24c of the current collector 24 are melted and compatible with each other to form a second seal portion 43. That is, a sealing member 4 is formed on the bipolar electrode 21. Similarly, each sealing member 4 is also formed on a positive terminal electrode 22 and a negative terminal electrode 23.

[0059] Next, as illustrated in FIG. 5, the bipolar electrode 21 with the sealing member 4 formed thereon, the positive terminal electrode 22 with the sealing member 4 formed thereon, the negative terminal electrode 23 with the sealing member 4 formed thereon, and each spacer 5 are stacked. Each spacer 5 is disposed between the corresponding sealing members 4. Next, as illustrated in FIG. 6, the respective outer side faces that the sealing members 4 and the spacers 5 have are melted by a heating device 8. The heating device 8 is, for example, an infrared heater. The heating device 8 irradiates the outer side faces of the sealing members 4 and the spacers 5 with infrared rays. As the outer side faces of the sealing members 4 and the spacers 5 are irradiated with the infrared rays, the respective temperatures of the sealing members 4 and the spacers 5 increase. The sealing members 4 and the spacers 5 are heated until the sealing member 4 and the spacer 5 are melted. As a result, the respective outer edges that the sealing members 4 and the spacers 5 have are melted. The melted outer edges of the sealing members 4 and the spacers 5 are solidified to form a welded portion 6. Next, an electrolytic solution is filled into each internal space S through the corresponding filling port. Next, the filling port is sealed. Accordingly, the power storage device 1 illustrated in FIG. 1 is manufactured.

[0060] As described above, in the power storage device 1, the current collector 24 includes the adhesive layer 243 that is provided between the first metal foil 241 and the second metal foil 242 and that bonds the first metal foil 241 and the second metal foil 242. The first seal portion 41 is welded to the first main face 24a of the current collector 24, and the second seal portion 43 is welded to the side face 24c of the current collector 24. Thus, heat in welding of the first seal portion 41 and the second seal portion 43 to the current collector 24 may be transferred to the adhesive layer 243. In particular, because the second seal portion 43 is in contact with the side face 243c of the adhesive layer 243, heat in welding of the second seal portion 43 to the current collector 24 is easily transferred to the adhesive layer 243. Here, the heat-resistant temperature of the adhesive layer 243 of the current collector 24 is higher than the respective melting points of the first seal portion 41 and the second seal portion 43. Thus, even if heat in welding of the first seal portion 41 and the second seal portion 43 to the current collector 24 is transferred to the adhesive layer 243, a decrease in adhesion strength by the adhesive layer 243 is inhibited. As a result, damage to the current collector 24 due to such a decrease in adhesion strength by the adhesive layer 243 is inhibited. Therefore, the power storage device 1 enables inhibition of damage to the bipolar electrode 21.

[0061] The power storage device 1 includes the spacers 5. Each spacer 5 is provided between such bipolar electrodes 21 as described above adjacent to each other in the stacking direction D, and is

welded to the corresponding sealing member **4**. The heat-resistant temperature of the adhesive layer **243** is higher than the melting point of the spacer **5**. As a result, even if heat in welding of the spacer **5** to the sealing member **4** is transferred to the adhesive layer **243**, a decrease in adhesion strength by the adhesive layer **243** is inhibited. Therefore, damage to the current collector **24** due to a decrease in adhesion strength by the adhesive layer **243** is inhibited, and as a result, damage to such a bipolar electrode **21** as described above is inhibited.

[0062] The sealing member **4** includes the third seal portion **42** welded to the second main face **24b** of the current collector **24**. The heat-resistant temperature of the adhesive layer **243** is higher than the melting point of the third seal portion **42**. Thus, even if heat in welding of the third seal portion **42** to the current collector **24** is transferred to the adhesive layer **243**, a decrease in adhesion strength by the adhesive layer **243** is inhibited. Therefore, damage to the current collector **24** due to a decrease in adhesion strength by the adhesive layer **243** is inhibited, and as a result, damage to the bipolar electrode **21** is inhibited.

[0063] The adhesive layer **243** contains the thermosetting resin. Thus, use of the thermosetting resin as a component of the adhesive layer **243** results in an increase of the heat-resistant temperature of the adhesive layer **243**. Such use of the thermosetting resin as a component of the adhesive layer **243** results in inhibition of deformation of the adhesive layer **243** due to heating, and results in inhibition of deformation of the current collector **24**. The sealing member **4** contains the thermoplastic resin. Thus, use of the thermoplastic resin as a component of the sealing member **4** results in enhancement of the adhesiveness of the sealing member **4** to the current collector **24** in welding of the sealing member **4** to the current collector **24**, and results in enhancement of sealing performance by the sealing member **4**.

[0064] The embodiments of the present disclosure are described above, but the present disclosure is not limited to the above-described embodiments.

[0065] As illustrated in FIG. 7, a current collector **24** may further include carbon coat layers **244** each provided between a first metal foil **241** and an adhesive layer **243** and between a second metal foil **242** and the adhesive layer **243**. Such a carbon coat layer **244** as described above is conductive. The carbon coat layer **244** contains, for example, a binder and graphite. The carbon coat layer **244** may be provided either between the first metal foil **241** and the adhesive layer **243** or between the second metal foil **242** and the adhesive layer **243**. That is, the carbon coat layer **244** may be provided at least either between the first metal foil **241** and the adhesive layer **243** or between the second metal foil **242** and the adhesive layer **243**. According to such a configuration, for example, as compared with a case where the first metal foil **241** or the second metal foil **242** directly bonded to the adhesive layer **243**, the binder resin contained in the carbon coat layer **244** is more firmly bonded by the adhesive layer **243**. Thus, the adhesion strength between the first metal foil **241** and the adhesive layer **243** or the adhesion strength between the second metal foil **242** and the adhesive layer **243** can be more reliably secured. Further, for example, even if pinholes are formed in the first metal foil **241**, the second metal foil **242**, or the adhesive layer **243**, the liquid junction between internal spaces **S** adjacent to each other can be inhibited by the carbon coat layer **244**. Note that the carbon coat layer **244** may be further provided on a first main face **24a** or a second main face **24b** that the current collector **24** has. In this case, the carbon coat layer **244** is also interposed between the first main face **24a** and a first seal portion **41** or between the second main face **24b** and a third seal portion **42**.

[0066] In the embodiments, the description has been given focusing on the single bipolar electrode **21**; however, the positive terminal electrode **22** or the negative terminal electrode **23** may be focused. When focusing on the negative terminal electrode **23**, the second main face **24b** of the current collector **24** of the negative terminal electrode **23** corresponds to a first main face, the first main face **24a** of the current collector **24** of the negative terminal electrode **23** corresponds to a second main face, the layer including the second main face **24b** of the current collector **24** corresponds to a first metal foil, and the layer including the first main face **24a** corresponds to a

second metal foil.

[0067] In the embodiments, an example in which the adhesive layer **243** contains the thermosetting resin has been described, but the adhesive layer **243** may contain a thermoplastic resin. In the case, the heat-resistant temperature of the adhesive layer **243** may be the melting point of the adhesive layer **243**.

[0068] In the embodiments, an example in which the peripheral edge of the separator **27** is located between the sealing member **4** and the spacer **5** has been described, but the peripheral edge of the separator **27** may not reach the spacer **5**. The peripheral edge of the separator **27** may be welded to the sealing member **4**.

[0069] The gist of the present disclosure is as [1] to [5] below. [0070] [1] A power storage device including: an electrode including: a current collector having a first main face; and an active material layer provided on the first main face; and a sealing member provided on a surface of the current collector, the sealing member surrounding the active material layer when viewed from a direction perpendicular to the first main face, in which the current collector includes: a first metal foil including the first main face; a second metal foil provided opposite to the active material layer with respect to the first metal foil; and an adhesive layer provided between the first metal foil and the second metal foil, the adhesive layer bonding the first metal foil and the second metal foil with each other, the sealing member includes: a first seal portion welded to the first main face; and a second seal portion welded to a side face of the current collector, the side face including a side face of the adhesive layer, and a heat-resistant temperature of the adhesive layer is higher than respective melting points of the first seal portion and the second seal portion. [0071] [2] The power storage device according to [1] further including a spacer, in which the electrode is each of a plurality of electrodes stacked in a stacking direction, the spacer is provided between the electrodes adjacent to each other in the stacking direction, the spacer being welded to the sealing member, and the heat-resistant temperature of the adhesive layer is higher than a melting point of the spacer. [0072] [3] The power storage device according to [1] or [2], in which the current collector has a second main face opposite to the first main face, the sealing member further includes a third seal portion welded to the second main face, and the heat-resistant temperature of the adhesive layer is higher than a melting point of the third seal portion. [0073] [4] The power storage device according to any of [1] to [3], in which the adhesive layer contains a thermosetting resin, and the sealing member contains a thermoplastic resin. [0074] [5] The power storage device according to any of [1] to [4], in which the current collector further includes a carbon coat layer provided at least either between the first metal foil and the adhesive layer or between the second metal foil and the adhesive layer.

REFERENCE SIGNS LIST

[0075] **1** Power storage device [0076] **4** Sealing member [0077] **5** Spacer [0078] **21** Bipolar electrode [0079] **22** Positive terminal electrode [0080] **23** Negative terminal electrode [0081] **24** Current collector [0082] **24a** First main face [0083] **24b** Second main face [0084] **25** Positive active material layer [0085] **26** Negative active material layer [0086] **41** First seal portion [0087] **42** Third seal portion [0088] **43** Second seal portion [0089] **241** First metal foil [0090] **242** Second metal foil [0091] **243** Adhesive layer [0092] **244** Carbon coat layer

Claims

1. A power storage device comprising: an electrode including a current collector having a first main face, and an active material layer provided on the first main face; and a sealing member provided on a surface of the current collector, the sealing member surrounding the active material layer when viewed from a direction perpendicular to the first main face, wherein the current collector includes a first metal foil including the first main face, a second metal foil provided opposite to the active material layer with respect to the first metal foil, and an adhesive layer provided between the first

metal foil and the second metal foil, the adhesive layer bonding the first metal foil and the second metal foil with each other, the sealing member includes a first seal portion welded to the first main face, and a second seal portion welded to a side face of the current collector including a side face of the adhesive layer, and a heat-resistant temperature of the adhesive layer is higher than respective melting points of the first seal portion and the second seal portion.

2. The power storage device according to claim 1 further comprising a spacer, wherein the electrode is each of a plurality of electrodes stacked in a stacking direction, the spacer is provided between the electrodes adjacent to each other in the stacking direction, the spacer being welded to the sealing member, and the heat-resistant temperature of the adhesive layer is higher than a melting point of the spacer.

3. The power storage device according to claim 1, wherein the current collector has a second main face opposite to the first main face, and the sealing member further includes a third seal portion welded to the second main face, and the heat-resistant temperature of the adhesive layer is higher than a melting point of the third seal portion.

4. The power storage device according to claim 1, wherein the adhesive layer contains a thermosetting resin, and the sealing member contains a thermoplastic resin.

5. The power storage device according to claim 1, wherein the current collector further includes a carbon coat layer provided at least either between the first metal foil and the adhesive layer or between the second metal foil and the adhesive layer.
